IOT AND CLOUD COMPUTING

Course	B.Tech.-VI-Sem.	L	T	P	C
Subject Code	22CAPC61	3	-	-	3

Course Outcomes (COs) & CO-PO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	Upon completion of course the students will be able to	PO2	PO3	PO6	PO7	PO12	PSO1
CO1	explain the concepts of IoT	3	2	3	3	3	3
CO2	illustrate the foundations of IoT	3	2	3	3	3	3
CO3	adapt protocol and standards of IoT	3	3	3	3	3	3
CO4	outline the importance of cloud in IoT	3	3	3	3	3	3
CO5	make use of cloud in IoT enabled spaces	3	2	3	3	3	3

Syllabus

Unit	Title/Topics	Hours
I	**Introduction**	10
Introduction to Internet of Things, IoT Kaleidoscope, Ubiquitous IoT Applications, A Panoramic View of IoT Applications, Telematics and Intelligent Transport Systems, Smart Grid and Electric Vehicles, Smarter Planet and Smart Buildings.		
II	**Pillars and DNA of IoT**	10
Four Pillars of IoT, M2M: The Internet of Devices, RFID: The Internet of Objects, WSN: The Internet of Transducers, SCADA: The Internet of Controllers, The DNA of IoT - DCM: Device - Things that Talk. Connect - Via Pervasive Networks, Wired Networks, Wireless Networks. Manage - To Create New Business Value.		
III	**Smart Home Scheduling and Cloud Computing**	4+5=9
Part-A: Protocol Standards for IoT: TCP and UDP, M2M and WSN Protocols, SCADA and RFID Protocols, Issues with IoT Standardization.		
Part-B: Architecture Standardization for WoT: Platform Middleware for WoT, Standards for M2M, Frameworks for WSN, Standards for SCADA, Extensions on RFID Standards, Unified Multitier WoT Architecture, OSGi: The Universal Middleware, WoT Framework Based on Data Standards.		
IV	**The Cloud of Things**	10
Introduction to Cloud Computing, Cloud Middleware, NIST's SPI Architecture and Cloud Standards, Cloud Providers and Systems, The Cloud of Things, The Internet of Things and Cloud Computing, Mobile Cloud Computing, MAI versus XaaS: The Long Tail and the Big Switch, The Cloud of Things Architecture, Four Deployment Models, Vertical Applications.		
V	**Cloud in IoT Enabled Spaces**	9
Medium Access, Data Caching, Smart Parking, Indecision Service Delivery, Home, Learning in Cities', Data Delivery Pricing, Planting & Farming.		
Textbooks		
1. H. Zhou, "The Internet of Things in the Cloud: A Middleware Perspective", CRC Press, 2012. 2. Maheswaran, Muthucumaru et.al., "The Cloud in IoT-enabled Spaces", CRC Press, 2019.		
References		
1. HwaiyuGeng, "Internet of Things and Data Analytics Handbook", Wiley, 2016. 2. Al-Turjman, Fadi, "Trends in Cloud-based IoT", Springer, 2020.		